

CSE331: Automata and Computability
Assignment 02

Group Formation: You may form a group of **at most 3 people**.

Deadlines and Submission Links:

- CFG (Question 3):
 - Deadline: Dec 12, 2025 11:59 PM
 - Submission Link: <https://forms.gle/tj8teURsi3yEiKaw6>
 - Submit the Hard copy also.

- Pumping Lemma (Questions 2) and Ambiguity (Questions 5):
 - Deadline: Dec 19, 2025 11:59 PM
 - Submission Link: <https://forms.gle/oZgZGXrENAu1H73C9>
 - Submit the Hard copy also.

- ~~Question 01, Pushdown Automata (Questions 4), Turing Machine (Questions 6):~~
 - Deadline: Dec 27, 2025 11:59 PM
 - Submission Link: <https://forms.gle/pKRmmbKZjnXkXyqZ9>
 - Submit the Hard copy also.

- Question 01 and Turing Machine (Questions 6):
 - Deadline: ~~Jan 7, 2026 11:59 PM~~
 - Jan 8, 2026 11:59 PM
 - Submission Link: <https://forms.gle/Uuak3hi1PoJUByEE9>

We will follow the deadline strictly. After the deadline ends, I will share the solution, so no submissions will be accepted after the solution is provided.

[This doc file contains both the Assignment and [Practice Sheet](#) [\[Click Here\]](#) on the final syllabus. The solution for the practice sheet can be found [\[Click Here\]](#) [Self Note: Link should be updated - Done]

Solve all the questions given below:

- 1) List five use cases where and how you can apply the knowledge from our CSE331 course.
- 2) Proof L is a non-regular language using the Pumping Lemma:
 - a) $L = \{w \in \{0,1\}^*: w = 0^{n!}, n \geq 0\}$
 - b) $L = \{w \in \{0,1\}^*: w = 0^n 1^n 1, n \geq 0\}$
 - c) $L = \{w \in \{0,1\}^*: w = 0^a 1^b 1^c 0^d, \text{ where } a + b = c + d \text{ and } a, b, c, d \geq 0\}$ [Be careful for (b): check if xyyz works]
 - d) $L = \{w \in \Sigma^*: w = a^i b^j, \text{ where } i > j, \text{ and } j \geq 0\}$
 - e)

3) Design a Context Free Grammar for the Language:

- a) $L = \{w \in \{a,b,c,p,q,r,\#\}^*: a^i \#^n c^j p^{2x} q^y r^z b^k \text{ where } i=j+k, y=3x+z, z \text{ is even, } n \text{ is odd, and } i,j,k,n,x,y,z \geq 0\}$
- b) $L = \{w \in \{0,1,2\}^*: w = 0^i 2^j 1^k, [\text{whereconditions.....}] \}$

where...

- i) $i = k, i, k \geq 1 \text{ and } j \geq 2$
- ii) $i = 3k, j \text{ is odd, and } i, j, k \geq 0$
- iii) $i \text{ is a multiple of two, } k \text{ is two more than a multiple of 3, } j = k+i, \text{ and } i, j, k \geq 0$
- iv) $i+j > k \text{ and } i, j, k \geq 0$
- v) $i+k \text{ is even, } j = i+k \text{ and } j \geq 1$
- c) $L = \{w \in \{0,1\}^*: \text{ the parity of 0s and 1s is different in } w\}$
- d) $L = \{w \in \{0,1\}^*: \text{ the number of 0s and 1s are different in } w\}$
[Hint: First, try to solve for an equal number of 0s and 1s in w]
- e) $L = \{w \in \{a, b\}^*: w \text{ is a palindrome}\}$. Design a CFG for $\bar{L}$
- f) $L = \{w \in \{a, b\}^*: w \text{ does not contain any palindromic substring of length two}\}$

g) $L = \{w\#x \mid w^R \text{ is a substring of } x \text{ where } w \in \{0, 1\}^*\}$.

w^R Denotes reverse of string, w .

h) $L = \{1^i 0 2^j 1^k \mid i, j, k \geq 0, 3i \geq 4k + 2, j \text{ is not divisible by three}\}$

i) Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{0, 1\}$

$L1 = \{w \in \Sigma^* : w \text{ contains exactly two 1s}\}$

$L2 = \{x\#y : x \in \Sigma^*, y \in L1, |x| = |y|\}$

Construct a CFG for $L2$.

j) Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{0, 1\}$

$L1 = \{w \in \Sigma^* : w \text{ contains at least three 1s}\}$

$L2 = \{x\#y : x \in (\Sigma\Sigma)^*, y \in L1, |x| = |y|\}$

Construct a CFG for $L2$.

k) Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{a, b\}$

$L1 = \{w \in \Sigma^* : w \text{ contains } bb \text{ as a substring}\}$

$L2 = \{w1\#w2 : w1, w2 \in L1 \text{ and } |w1| = |w2|\}$

Construct a CFG for $L2$.

4) For all the problems in Question(3), now construct the Pushdown Automata (PDA).

5) Derivation, Parse Tree, Ambiguity

i) Given the Context-Free Grammar, answer the following questions

$$A \rightarrow A1 \mid 0A1 \mid 01$$

- (a) Give a leftmost derivation for the string 001111.
- (b) Sketch the parse tree corresponding to the derivation you gave in (a).
- (c) Demonstrate that there are two more parse trees (apart from the one you already found in (b)) for the same string.
- (d) Find a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)

ii) Given the Context-Free Grammar, answer the following questions:

$$\begin{aligned}A &\rightarrow 1A \mid 1C \mid 0B \mid 00A \\B &\rightarrow 0A \mid 1B \mid 00B \\C &\rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \varepsilon\end{aligned}$$

- (a) Give a leftmost derivation for the string 01011001. (3 points)
- (b) Sketch the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. (3 points)
- (d) Find a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)
- (e) Design an unambiguous Context Free Grammar for the language represented by the given ambiguous grammar. (1 point)

6) Turing Machine

- a) $L = \{1^{3^n} \mid n \geq 0\}$
- b) $L = \{w \in \{0,1,2\}^*: 0^i 1^j 2^k \text{ where } i=j=k \text{ and } i,j,k \geq 0\}$
- c) $L = \{w \in \{0,1,2\}^*: 0^i \# 1^j \# 2^k \text{ where } i*j=k \text{ and } i,j,k \geq 0\}$ [Idea described in Sipser's Book]
- d) $L = \{w \in \{a,b\}^*: w \text{ is an odd length palindrome}\}$
- e) $L = \{w \# w^R \# w \mid w \in \{a,b\}^*\}$ [w^R means reverse of w]
- f) Let $\text{HALT_TM} = \{ \langle M, w \rangle : M \text{ is a TM and } M \text{ halts on input } w \}$. Given that HALT_TM is undecidable, prove that the collection of Turing-recognizable languages is not closed under complement.
- g)

CSE331: Automata and Computability Practice Sheet Prepared By: KKP

The solution for the practice sheet can be found [\[Click Here\]](#)
[Self Note: Link should be updated - Done]

[There are also some practice problems in the handnotes]

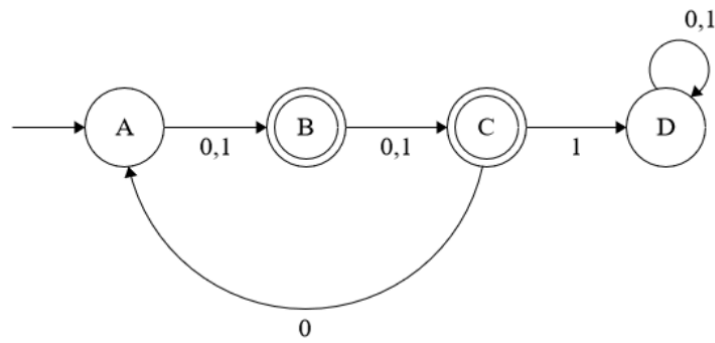
Question 1: Designing Context-Free Grammar

Give Context-free Grammar that generates the language

- a) $L = \{w \in \{0,1\}^* : w \text{ starts with '0' and the length of } w \text{ is even.}\}$
- b) $L = \{w \in \{a,b\}^* : \text{every second letter in } w \text{ is a 'b'.}\}$
- c) $L = \{w \in \{a,b\}^* : \text{the length of } w \text{ is divisible by three}\}$
- d) $L = \{w \in \{0,1\}^* : w \text{ starts with and ends with a different symbol.}\}$
- e) $L = \{w \in \{a,b\}^* : \text{the number of 'a' is at least the number of 'b' in } w.\}$
[Hint: [Link](#)]
- f) $L = \{w \in \{a,b\}^* : \text{the length of } w \text{ is two more than a multiple of six}\}$
- g) $L = \{w \in \{0,1\}^* : w = 0^n 1^{2n+3}, n \geq 0\}$
- h) $L = \{w \in \{0,1\}^* : w = 0^n 1^n, \text{ where } n \text{ is odd.}\}$
- i) $L = \{w \in \{0,1,2\}^* : w = 0^i 1^j 2^k \text{ where } j \geq 2i + 3k, \text{ and } i, k \geq 0\}$
- j) $L = \{w \in \{0,1,2,3\}^* : w = 0^i 1^j 2^k 3^m \text{ where } i=m, j \geq 3k+2, \text{ and } m, k \geq 0\}$
- k) $L = \{w \in \{0,1,2\}^* : w = 0^i 1^j 2^k \text{ where } i > 2j + 3k, \text{ and } j, k \geq 0\}$
- l) If $A = \{w \in \{0, 1\}^* : w \text{ contains at least two 0s}\}$, then construct $L = \{w \in \{0, 1\}^* : w = 0^{3i} v 1^{2i} \text{ where } v \in A \text{ and } i \geq 0\}$ [marked as problem j in the solution]
- m) Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{0,1\}$
 $L_1 = \{w \in \Sigma^* : w \text{ contains } 11\}$
 $L_2 = \{x\#y : y \in L_1, x \in \Sigma^*, |x| = |y|\}$
 Construct a CFG for L_2 .
- n) $L = \{w \in \{0,1\}^* : w_1\#w_2 : \text{number of 0s } w_1 \text{ is equal to number of 1s in } w_2.\}$
- o) $L = \{w \in \{0,1\}^* : w_1\#w_2 : \text{length of } w_2 \text{ is double of length of } w_1.\}$

Construct a Context-free Grammar for the language expressed by the following Finite Automata:

- a) Convert the Regular Expression into a CFG: $((ab)^* + (a+b^3)cb)^*$
- b) Convert the Regular Expression into a CFG: $(a^*b^* + (ac+b^*c)b)^*$
- c) Convert the following DFA into a CFG:



Question 2: Derivation, Parse tree, Ambiguity

I.

$$S \rightarrow ASB \mid SS \mid SAS \mid A$$

$$A \rightarrow ASS \mid BS \mid B$$

$$B \rightarrow 00 \mid 11 \mid 01 \mid 1$$

Given the above Context-Free Grammar, answer the following questions:

- Show the Leftmost Derivation of 00010111
- Sketch the parse tree corresponding to the derivation you gave in (a).
- Show the Rightmost Derivation of 00010111
- Sketch the parse tree corresponding to the derivation you gave in (c).

II. $S \rightarrow 0S1 \mid 1S0 \mid SS \mid 01 \mid 10$

Given the above Context-Free Grammar, answer the following questions:

- Show that the grammar above is ambiguous by demonstrating two different parse trees for 011010.
- Find a string w of length six such that w has exactly one parse tree in the grammar above.

Question 3: Pushdown Automata

a) $L = \{ w \in \{0,1\}^* : w = 0^n 1^n, \text{ where } n \text{ is odd.} \}$

b) $L = \{ w \in \{0,1\}^* : w = 0^m 1^n, \text{ where } m, n \geq 1 \text{ and } m \geq n. \}$

- c) $L = \{ w \in \{0,1\}^* : w \mid \text{the length of } w \text{ is divisible by four} \}$
- d) $L_1 = \{ w \in \{0,1\}^* : \text{the number of 1s in } w \text{ is multiple of 3.} \}$
 $L_2 = \{ w \in \{0,1\}^* : w \text{ contains even numbers of 0.} \}$
Construct a PDA for $L = \{ w \in \{0,1\}^* : w = uv, \text{ where } u \in L_1, v \in L_2 \text{ and } |u| = |v| \}$
- e) $L = \{ w \in \{0,1\}^* : w = 0^i 1^j 0^k, \text{ where } j = i+k \text{ and } i, k \geq 0. \}$
- f) $L = \{ w \in \{0,1\}^* : w = 0^i 1^j 0^k, \text{ where } i+j = k \text{ and } i, k \geq 0. \}$
- g) $L = \{ w \in \{0,1\}^* : w_1 \# w_2 : \text{number of 00 substrings in } w_1 \text{ is equal to number of 11 in } w_2. \}$
- h) $L = \{ w \overline{w}^R : w \in \{0,1\}^* \}$

$$L = \{ w \overline{w}^R : w \in \{0,1\}^* \}$$

Here, $\overline{w}^R$ denotes the reverse complement of the string w .

For example, $01001101 \in L$, because the second half of the string, 1101 , is the reverse complement of the first half, 0100 , i.e. $1101 = \overline{0100}^R$

Question 4: Chomsky Normal Form - CNF [Not included in the Final Syllabus]

- a) Convert the following grammar into Chomsky Normal Form. You must show work.

$$\begin{aligned} S &\rightarrow bBS \mid \varepsilon \\ B &\rightarrow bYb \mid bY \\ C &\rightarrow Ccb \mid c \mid A \\ Y &\rightarrow bcY \mid \varepsilon \mid YBF \end{aligned}$$

Here b , and c are terminals and the rest are variables.

- b) Convert the following grammar into Chomsky Normal Form. You must show work.

$$\begin{aligned} S &\rightarrow YaSb \\ X &\rightarrow aXY \mid bX \mid Y \\ Y &\rightarrow X \mid b \mid \varepsilon \end{aligned}$$

Here a , and b are terminals and the rest are variables.

- c) Convert the following grammar into Chomsky Normal Form. You must show work.

$$\begin{aligned} S &\rightarrow bXaY \mid ZXb \\ X &\rightarrow aY \mid bY \mid Y \\ Y &\rightarrow X \mid c \mid \varepsilon \end{aligned}$$

$$Z \rightarrow ZaX$$

Here a, b, c are terminals and the rest are variables.

d)

Problem 3 (CO2): Chomsky Normal Form (10 points)

- (a) **List** the rules that violate the conditions of Chomsky Normal form in the following grammar. Here a, b, and c are terminals and the rest are variables.

$$\begin{aligned} A &\rightarrow BC \mid bB \mid a \\ B &\rightarrow bb \mid Cb \mid b \mid C \\ C &\rightarrow c \end{aligned}$$

- (b) **Write** down the additional rules that need to be added to the following grammar if the production, $B \rightarrow \varepsilon$ is removed. Here 0 and 1 are terminals and the rest are variables.

$$\begin{aligned} S &\rightarrow AB \mid 1 \\ A &\rightarrow BAB \mid ABA \mid B \mid 11 \\ B &\rightarrow 00 \mid \varepsilon \end{aligned}$$

- (c) **Write** down the additional rules that need to be added to the following grammar if the unit productions are removed. Here 0 and 1 are terminals and the rest are variables.

$$\begin{aligned} S &\rightarrow XYX \mid YX \mid X \mid Y \\ Y &\rightarrow XY \mid X0 \mid 0 \\ X &\rightarrow 1 \mid Y \end{aligned}$$

Question 5: CYK Algorithm [Not included in the Final Syllabus]

- a) Consider the following grammar in CNF form.

$$\begin{aligned} S &\rightarrow BC \mid CD \\ B &\rightarrow CB \mid c \\ C &\rightarrow DD \mid c \\ D &\rightarrow BC \mid b \end{aligned}$$

Show if the string $w=cccbb$ can be derived from the grammar above.

- b) Apply the CYK algorithm to determine whether the string "abcaa" can be derived in the following grammar. You must show the entire CYK table. Here a, b, c are terminals and the rest are variables.

$$\begin{aligned} S &\rightarrow CA \\ A &\rightarrow AA \mid AD \mid a \\ B &\rightarrow AB \mid BC \mid b \\ C &\rightarrow CA \mid BC \mid c \end{aligned}$$

$$D \rightarrow a$$